
Erosion of Lizard Diversity by Climate Change and Altered Thermal Niches

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Source: *Science*, 14 May 2010, New Series, Vol. 328, No. 5980 (14 May 2010), pp. 894-899

Published by: American Association for the Advancement of Science

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could occur in *C. maculatus* through sexual selection on males (18, 26–28). If sexual selection is responsible for the greater strength of the r_{ID-BV} coefficients in males, it raises the possibility of positive feedback, where sexual selection increases the contribution of deleterious mutations to trait expression, in turn increasing both good genes benefits from sexual selection and the benefit of sex itself.

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29. We thank J. Chan for the maintenance of the pedigree and the collection and management of the data set; R. Black and F. Gonzalez for statistical advice; B. Booth, A. Sutton, and S. Jennings for their assistance with the experiment; and N. Colegrave, W. Hazel, M. Puurtinen, J. Radwan, M. Ritchie, and L. Simmons for their comments on the manuscript. This work was supported by Australian Research Council fellowships to J.L.T. and N.R.L.

Supporting Online Material

www.sciencemag.org/cgi/content/full/328/5980/892/DC1

Materials and Methods

SOM Text

Figs. S1 to S6

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5 February 2010; accepted 24 March 2010

10.1126/science.1188013

Erosion of Lizard Diversity by Climate Change and Altered Thermal Niches

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It is predicted that climate change will cause species extinctions and distributional shifts in coming decades, but data to validate these predictions are relatively scarce. Here, we compare recent and historical surveys for 48 Mexican lizard species at 200 sites. Since 1975, 12% of local populations have gone extinct. We verified physiological models of extinction risk with observed local extinctions and extended projections worldwide. Since 1975, we estimate that 4% of local populations have gone extinct worldwide, but by 2080 local extinctions are projected to reach 39% worldwide, and species extinctions may reach 20%. Global extinction projections were validated with local extinctions observed from 1975 to 2009 for regional biotas on four other continents, suggesting that lizards have already crossed a threshold for extinctions caused by climate change.

Global climate change affects organisms in all biomes and ecosystems. Two natural compensatory responses are possible. Given enough time and dispersal, species may shift to more favorable thermal environments, or they may adjust to new environments by behavioral plasticity, physiological plasticity, or adaptation. Alternatively, failure to adjust or adapt culminates in demographic collapse and extinction. Despite accumulating evidence of contemporary climate change affecting species ranges and phenologies (1–3), evidence of extinctions at either local or global scales is lacking (4–6). Moreover, current forecasting models (7, 8) are not calibrated with actual extinctions, but are premised on hypothesized effects of thermal physiology on demography and extinction. Alternatively, models are based on range shifts or species-area relations in mobile species

(1), but not extinctions (9). Hence, there is still much uncertainty regarding the expected magnitude of extinctions resulting from climate change (10).

Empirical validation of global extinction forecasts requires three forms of evidence. First, actual extinctions should be linked to macroclimate and validated to biophysical thermal causes arising from microclimate (11). Second, the pace of climate change should compromise thermal adaptation (10), such that evolutionary rates lag behind global warming owing to constraints on thermal physiology (12, 13). Third, extinctions due to climate should be global in extent.

From 2006 to 2008, we resurveyed 48 *Sceloporus* lizard species at 200 sites in Mexico that were first sampled in 1975 to 1995, and 12% of sites were locally extinct by 2009 (table S1).

Although *Sceloporus* lizards are heliotherms that bask and require solar radiation to attain physiologically active body temperatures (T_b) (14, 15), activity in hot weather may result in T_b exceeding CT_{max} , the critical thermal maximum, leading to death. Lizards retreat to cool refuges rather than risk death by overheating. However, hours of restriction (h_r) in thermal refuges limit foraging, constraining costly metabolic functions like growth, maintenance, and reproduction, thereby undermining population growth rates and raising extinction risk. Lizards could evolve higher T_b , but this brings them closer to CT_{max} , which increases risk of overheating. Extinction risk may increase because of other thermal adaptations. For example, viviparity, which is posited to be a thermal adaptation to cold climates (16), may elevate extinction risk because high T_b can compromise embryonic development in utero (17).

We analyzed rate of change in maximum air temperature T_{max} at 99 Mexican weather stations and constructed climate surfaces (tables S2 and S3, 1973 to 2008; fig. S1). Rate of change in T_{max} was greatest for winter-spring (January to May; fig. S1 and table S3A) and increased faster in northern and central México and at high elevation, as evidenced by significant coefficients for fitted climate surfaces. We found a correlation between rate of change in T_{max} during winter-spring breeding periods and local extinctions of *Sceloporus* species (table S3).

Many viviparous species in México are confined to high-elevation “islands,” where climate change has been most rapid. Logistic regression and multiple regression with phylogenetic independent contrasts (18, 19) revealed that extinction risk was significantly related to low latitudinal and altitudinal range limits (Fig. 1, A and B), where thermal physiology and/or ecological interactions limit species (20, 21). Phylogenetic correlation analysis (18) showed that extinction

risk of viviparous lizards (18%) was twice that of oviparous lizards (9%, $n = 10000$ bootstrap replications $P < 0.001$). Moreover, multiple regression based on phylogenetic independent contrasts (PICs; Fig. 1C and table S4) showed that extinction risk of viviparous taxa was significantly related to low $\bar{T}_b$ ($\bar{T}_{b,viviparous} = 31.8^\circ\text{C} \pm 0.31$ [SE], $\bar{T}_{b,oviparous} = 34.8^\circ\text{C} \pm 0.40$, $t = 5.92$, $P < 0.0001$) and cool montane habitats ($\bar{T}_{air,viviparous} = 22.4^\circ\text{C} \pm 1.79$, $\bar{T}_{air,oviparous} = 28.39^\circ\text{C} \pm 1.38$, $t = 2.89$, $P < 0.006$), where climate has changed most rapidly in México.

To validate patterns of extinction risk and T_b , we deployed thermal models (22) that record operative temperatures (T_e) at two extinct and two persistent Yucatán sites of *S. serrifer*. Hours of restriction in activity (h_r) during reproduction was significantly higher at extinct versus persistent sites ($t = 9.26$, $P < 0.0001$). By April 2009, h_r at extinct Yucatán sites had become so severe that if *S. serrifer* were still present, it would have to retreat shortly after emergence (fig. S4A). Daily T_{max} was positively correlated with h_r assessed by T_e ($P < 0.001$, fig. S4B). The relation between h_r as a function of T_{max} relative to *S. serrifer*'s T_b [$h_r = 6.12 + 0.74 \times (T_{max} - T_b)$,

eq. S2 (23)] is a general formula for predicting extinctions.

We modeled extinct/persistence status based on values for h_r at *Sceloporus* sites derived from eq. S2 (23). The Yucatán ground truth for *S. serrifer* suggests that extinction occurs when h_r exceeds 4. We calibrated this value with extinct/persistent *Sceloporus* sites. Goodness-of-fit tests of the model indicate that the best fit for observed and predicted extinctions at *Sceloporus* sites is $h_r > 3.85$. If a species with a given T_b at a given geo-referenced site, subjected to $T_{max,i}$, experienced $h_r > 3.85$ during the 2-month reproductive period (March to April), we assumed that it would go extinct by 2009. Association of predicted and observed extinctions from this physiological model was significant for oviparous ($\chi^2 = 49.0$, $P < 0.001$) and viviparous taxa ($\chi^2 = 4.2$, $P < 0.04$).

As demography of high-elevation taxa becomes compromised due to climate change, species at low elevation that were previously limited by physiology and competition should expand into historically cooler habitat that is now warmer (20, 24), perhaps accelerating extinction of high-elevation forms. For viviparous taxa, six erroneously assigned extinct sites involved six of the eight cases of range expansion by low-elevation taxa, which all invaded from low to high altitudes or latitudes (table S1; significant by sign test, $P < 0.001$). Adding range shifts of competitors as a factor improved fit significantly between observed and predicted extinctions ($\Delta\log\text{likelihood} = 45.37$, 1 df, $P < 0.0001$, logistic regression). Therefore, competitive exclusion by invading low-elevation taxa appears to exacerbate climate-change extinctions of high-elevation taxa.

Lizards cannot evolve rapidly enough to track current climate change because of constraints arising from the genetic architecture of thermal preference (12, 13). A phylogenetic correlation between T_b and CT_{max} constrains adaptation. PIC regression of CT_{max} on T_b among Phrynosomatidae suggests that a shift in T_b by 1°C yields only a 0.5°C correlated response in CT_{max} (table S5 and fig. S7). Thus, CT_{max} may not evolve fast enough to keep up with evolved change in T_b . Furthermore, adaptive increase in T_b due to climate change is constrained by genetic correlations in which high T_b necessarily requires prolonged activity out of retreat sites (25), further increasing risk of overheating. Genetic trade-offs with energetically costly traits such as growth (25) also constrain adaptation.

The evolutionary response ($R = h^2s$; s is the selection differential) necessary to keep pace with climate change is further constrained by low heritability for T_b , which we previously estimated at $h^2 = 0.17$ for *Sceloporus occidentalis* in the laboratory (25). We used the physiological model to compute the sustained selection differential at each site j , such that $T_{b,j} + \Delta T_{b,j}$ evolves to match $T_{max,j} + \Delta T_{max,j}$, yielding $\Delta h_{r,j} = 0$ and thereby rescuing population j from extinction [Δ , computed over 1975 to 2009 (historical), 2009 to

2050, and 2050 to 2080]. We assumed $s_j = R_j/h^2 = \Delta T_{b,j}/h^2$, and generation times of 1 year versus 2 years (i.e., lowland versus montane *Sceloporus*, table S1). We expressed these critical levels of adaptive response as surfaces for $s_{sustained}$, the sustained selection differential (Fig. 2B).

We compared the magnitude of selection allowing a species to adapt to climate change with maximum rates sustained under artificial or natural selection (26). Such comparisons are facilitated by dividing each sustained selection differential by the standard deviation ($\sigma_{T_b} = 1.23$ for T_b of Mexican lizards) to obtain i , the standardized intensity of selection (26). Whereas $i > 0.4$ can be sustained in laboratory artificial selection for nine generations (27), studies in nature (26) indicate that $i > 0.4$ computed on an annual basis are rare (<5%). We also reference i to other anthropogenic causes of selection. Overfishing of Atlantic cod yielded $i = 0.55$, among the highest measured, but this selection regime caused demographic collapse of the fishery (28). In México, extinct sites sustained significantly higher i than persistent sites ($i_{extinct} = 0.34 \pm 0.05$ versus $i_{persistent} = 0.13 \pm 0.02$, $t = 4.17$, $P < 0.001$). The relation between intensity of selection and demographic collapse is simple. If sustained for decades, the mortality fraction necessary for selective shifts to new optima compromises population growth rate precipitating local extinction.

If climate change T_{max} continues unabated in México, 56% of viviparous sites will be extinct by 2050 and 66% by 2080 (Fig. 2B). For oviparous sites, 46% will be extinct by 2050 and 61% by 2080. Based on local extinction of all populations surveyed for species, we project 58% species extinction of Mexican *Sceloporus* by 2080. Species extinction (58% by 2080) mirrors local population extinction (61 to 66%) because high-elevation endemics will go completely extinct as widespread lowland taxa expand to high elevations.

We used the model to derive global extinction projections (Fig. 3) for 34 lizard families (Table 1) with 1216 geo-referenced T_b records (table S6). Our data include heliotherms that bask and thermoconformers that do not bask, but track ambient air and surface temperature. T_{max} was obtained from the WorldClim database (29) at 10-arc min resolution (1975, 2020, 2050, and 2080). We used distributional limits of heliothermic lizards of the world in 1975 to calibrate h_r by family, which if exceeded at a given site would precipitate extinction. The extinction model is easily adapted to thermoconformers that maintain T_b close to T_{air} or retreat when $T_{air} > T_{preferred}$. Assuming a sine wave for T_{air} between T_{min} and T_{max} (24-hour period), if the cumulative hours that $T_{air} > T_b$ for a thermoconformer at a given geo-referenced site (table S6) exceeded the h_r of a given lizard family, we assumed it would go extinct. Given $T_{max} - T_b$ at each geo-referenced site, we computed the h_r each species sustained in 1975, and for each family we used

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the upper 95% confidence level of h_r (Table 1) as the extinction threshold (iteratively estimated, given global climate surfaces). Calibration with

these 1975 distributional limits for *Sceloporus* yields $h_r = 3.9$, which was cross-validated by $h_r = 3.85$ computed from observed extinctions

in México (1975 to 2009), and $h_r = 4$, which was estimated directly from T_e at persistent *S. serrifer* sites on the verge of extinction.

Fig. 1. (A) Logistic regression of extinction probability (0 = extant, 1 = extinct) of *Sceloporus* lizards and reproductive mode: $\chi^2 = 7.41$, $P = 0.025$, Δ elevation ($\chi^2 = 8.53$, $P = 0.014$), Δ latitude ($\chi^2 = 7.14$, $P = 0.004$), and Δ longitude (not significant), where Δ refers to deviations from species range midpoints. (B) Phylogenetic independent contrasts (PICs) of lineage survival (survival probability of local populations) and Δ elevation ($t = 2.15$, $P = 0.03$), Δ latitude ($t = 3.94$, $P = 0.0001$), and Δ longitude ($t = 2.66$, $P = 0.009$). (C) PICs of lineage survival, T_b ($t = 2.32$, $P = 0.02$), T_{air} ($t = 2.31$, $P = 0.02$), and reproductive mode ($t = -2.92$, $P = 0.005$).

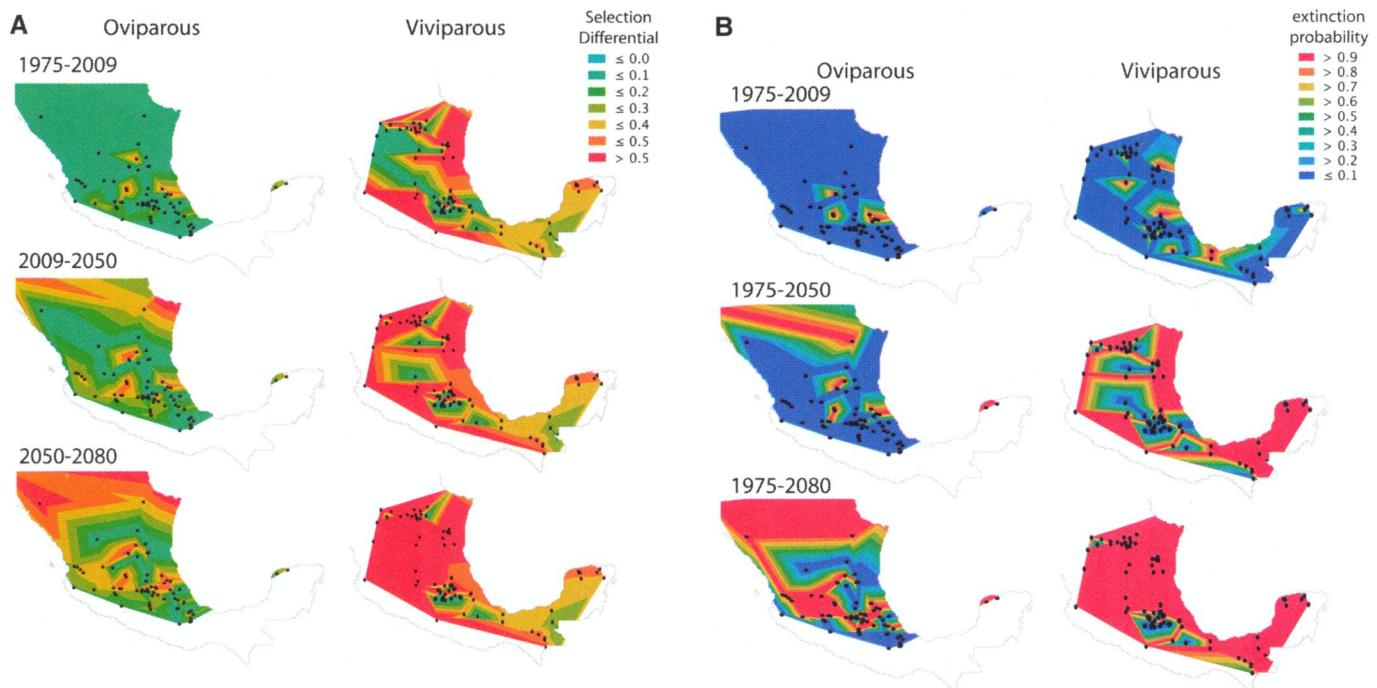
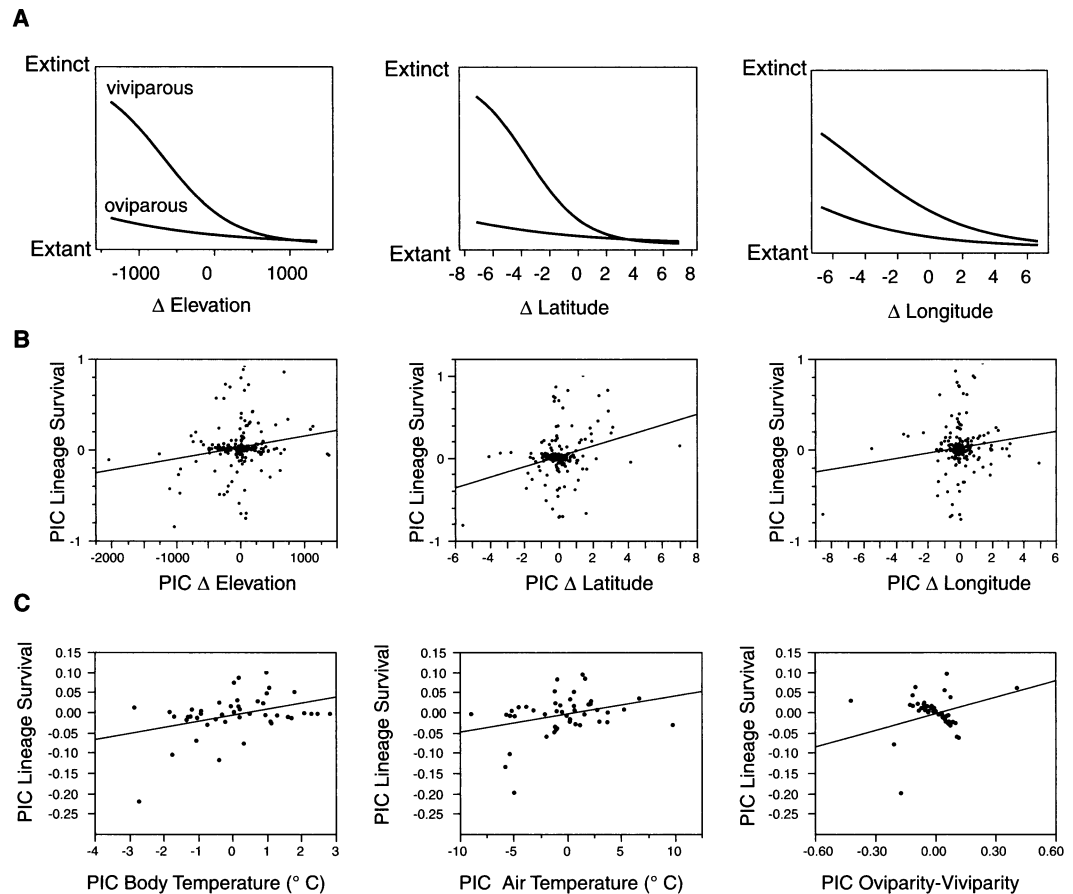


Fig. 2. (A) Sustained selection differentials per year required for T_b to keep pace with global warming. (B) Extinctions of Mexican *Sceloporus* lizards (1975 to 2009, 2009 to 2050, 2050 to 2080).

As in the validation of Mexican *Sceloporus* extinction, we computed h_r for temperate lizards over 2 critical reproductive months, but were conservative in modeling critical months required for h_r to be exceeded in the equatorial zone ($\pm 12^\circ$ latitude) where lizards potentially breed year-round (h_r exceeded over 12 months),

and in the wet-dry tropical zone ($\pm 12^\circ$ to 24° latitude: h_r exceeded for 5 to 6 months).

Geo-referenced T_b samples indicate that current (2009) local extinctions average 4% worldwide (Table 1). Global averages will increase fourfold to 16% by 2050 and nearly eightfold to 30% by 2080, while equatorial extinctions will reach 23%

by 2050 and 40% by 2080. Assuming reproduction shifts 1 month earlier in temperate zones [$h^2 = 1.0$ lay date (30)] and proportionately less to the trade zones (i.e., no shift), 2080 global extinctions jump to 38% because spring seasons are warming faster across the globe. Our model is robust to plasticity in T_b (table S7) and initial assump-

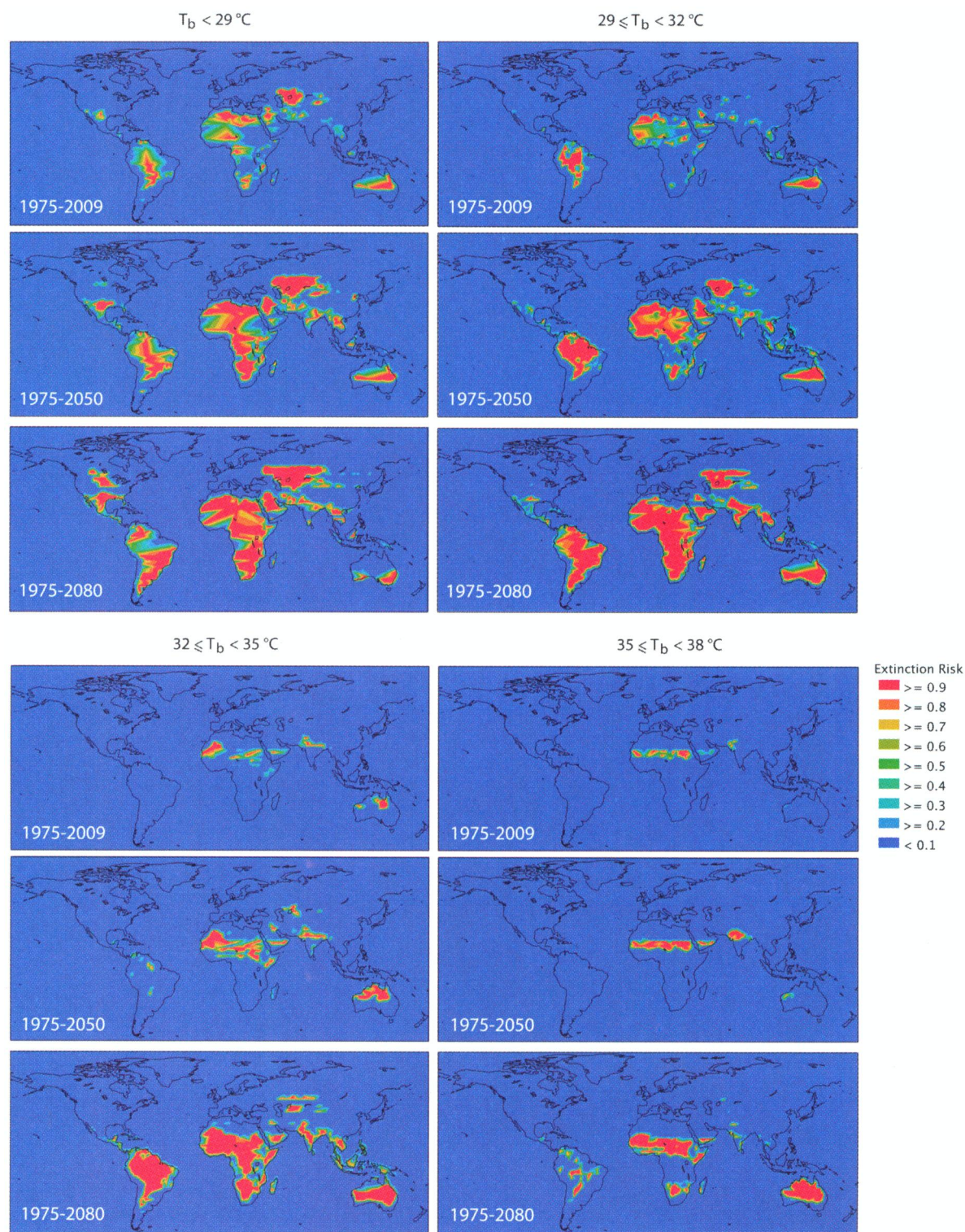


Fig. 3. Contour plots of global levels of local extinction for heliothermic lizards (1975 to 2009, 1975 to 2050, 1975 to 2080), assuming $\bar{h}_r = 4.55$ (23) and various T_b values.

Table 1. Sample size, T_b range, $\bar{T}_b \pm SE$, $\bar{T}_{max}$, h_r , and $n_{species}$ for 34 lizard families. Local extinction rates are based on geo-referenced T_b data and a physiological model of extinction. We also validated model predictions of local extinction risk in 2080 for six families: 57% (± 3 , $n = 200$) for Mexican Phrynosomatidae, 13% (± 2 , $n = 3155$) for South American Liolaemidae, 56%

(± 5 , $n = 117$) for European Lacertidae (*L. vivipara*), 13% (± 2 , $n = 1438$) for African Cordylidae + Gerrhosauridae, 57% (± 4 , $n = 125$) on Madagascar, and 10% (± 1 , $n = 2841$) for Australian Egernia Group lizards species. Estimates of species extinctions in each family are derived from the relationships for extinction of all local populations for these six families (table S8).

Family	<i>n</i>	<i>T</i> _b range	$\bar{T}_b \pm SE$	$\bar{T}_{max}$	<i>h_r</i>	Mode of thermoregulation	Local extinction levels				Species extinction	
							<i>n</i> _{spp}	2009	2050	2080	2050	2080
Agamidae	74	19.0–43.8	35.6±0.34	29.8	7.0	Heliothermic	381	0.000	0.169	0.292	0.059	0.266
Amphisbaenidae	2	21.1–21.2	21.2±0.05	28.8	16.2	Fossorial	160	0.000	0.000	0.250	0.000	0.228
						thermoconformer						
Anguidae	10	21.4–32.3	26.7±0.94	20.9	5.6	Heliothermic, a few fossorial	112	0.111	0.111	0.111	0.039	0.101
Annielliidae	2	21.0–23.6	22.3±2.09	20.5	11.5	Fossorial	2	0.000	0.000	0.000	0.000	0.000
						thermoconformer						
Chamaeleonidae	18	22.2–33.5	30.0±0.70	26.8	12.0	Forest	161	0.063	0.063	0.063	0.022	0.057
						thermoconformer						
Cordylidae	11	27.8–33.8	31.5±0.82	23.6	6.8	Heliothermic	54	0.000	0.000	0.200	0.000	0.182
Corytophanidae	4	26.0–35.0	31.9±1.48	29.4	13.4	Forest	9	0.250	0.250	0.250	0.088	0.228
						thermoconformer						
Crotaphytidae	23	35.5–38.9	37.3±0.62	23.3	1.2	Heliothermic	12	0.111	0.167	0.222	0.059	0.202
Carphodactylidae	11	15.1–35.5	24.5±1.59	34.7	10.9	Thermoconformer	30	0.350	0.820	0.820	0.289	0.748
Diplodactylidae	42	16.9–35.9	27.3±0.59	31.2	10.9	Thermoconformer	141	0.070	0.190	0.190	0.067	0.173
Eublepharidae	18	26.6–33.0	28.5±0.44	32.9	10.9	Thermoconformer	28	0.060	0.240	0.240	0.084	0.219
Gekkonidae	40	26.0–35.3	30.1±0.58	32.6	10.9	Thermoconformer	700	0.000	0.000	0.000	0.000	0.000
Phyllodactylidae	13	16.6–38.9	30.6±1.42	30.4	10.9	Thermoconformer	100	0.000	0.000	0.000	0.000	0.000
Pygopodidae	21	24.9–35.1	25.4±0.46	17.9	11.5	Fossorial	38	0.000	0.000	0.000	0.000	0.000
						thermoconformer						
Sphaerodactylidae	19	25.3–38.6	30.2±0.75	33.0	10.9	Thermoconformer	200	0.000	0.000	0.000	0.000	0.000
Gerrhosauridae	4	31.8–33.3	32.6±2.09	28.3	6.8	Heliothermic	16	0.333	0.333	0.333	0.117	0.304
Gymnophthalmidae	20	21.5–29.9	26.4±0.66	30.3	13.8	Leaf litter	193	0.095	0.333	0.667	0.117	0.608
						thermoconformer						
Helodermatidae	2	29.4–30.2	29.8±2.09	24.8	2.7	Heliothermic/thermal inertia	2	0.000	0.000	1.000	0.000	0.912
Iguanidae	20	32.9–42.1	37.3±0.79	28.1	3.7	Heliothermic	36	0.143	0.143	0.286	0.050	0.261
Lacertidae	89	26.7–40.2	35.4±0.31	25.6	3.1	Heliothermic	279	0.034	0.241	0.460	0.085	0.420
Lanthanotidae	1	–	28.0	30.5	9.4	Forest	1	1.000	1.000	1.000	0.352	0.912
						thermoconformer						
Leiocephalidae	1	–	36.3	31.7	2.8	Heliothermic	29	0.000	1.000	1.000	0.352	0.912
Liolaemidae	125	24.4–40.8	33.7±0.27	17.8	1.4	Heliothermic	219	0.027	0.071	0.107	0.025	0.098
Opluridae	3	36.2–39.8	37.7±1.71	31.8	4.0	Heliothermic	7	0.333	0.667	0.667	0.235	0.608
Phrynosomatidae	215	26.8–41.5	35.2±0.20	24.9	3.9	Heliothermic	125	0.037	0.087	0.149	0.031	0.136
Polychrotidae	121	19.6–35.0	29.6±0.27	29.6	14.4	Forest	393	0.018	0.043	0.068	0.015	0.062
						thermoconformer						
Scincidae	210	20.3–38.0	32.9±0.20	26.5	6.2	Heliothermic, a few fossorial	1305	0.015	0.092	0.308	0.032	0.281
Sphenodontidae	1	14.5–21.0	14.5±2.09	18.0	10.7	Nocturnal	1	0.000	0.000	0.000	0.000	0.000
						thermoconformer						
Teliidae	91	26.8–41.3	37.9±0.31	29.0	4.2	Heliothermic	121	0.012	0.136	0.210	0.048	0.192
Trogonophidae	2	22.0–22.5	22.3±0.25	28.8	16.2	Fossorial	8	0.000	0.000	0.000	0.000	0.000
						thermoconformer						
Tropiduridae	72	26.2–38.0	33.7±0.35	28.3	7.7	Heliothermic	111	0.043	0.058	0.087	0.020	0.079
Varanidae	46	28.8–38.9	35.8±0.44	29.7	4.6	Heliothermic/thermal inertia	68	0.001	0.023	0.178	0.008	0.162
Xantusidae	8	18.7–33.0	25.4±1.32	20.7	0.0	Thigmothermic	29	0.000	0.000	0.000	0.000	0.000
						thermoconformer						
Xenosauridae	5	20.3–25.6	23.2±1.48	26.4	11.4	Thigmothermic	6	0.200	0.200	0.600	0.070	0.547
						thermoconformer						

tions made for reproductive periods in the tropics. If h_r for equatorial taxa is computed over the 9 hottest months of reproduction, rather than the conservative assumption of 12 months, global extinctions increase to 39% by 2080.

The global generality of our model is verified by concordant distributions of current observed and predicted local extinctions of lizard biotas from four other continents (table S7). Our model pinpoints exact locations of two Liolaemid species

going extinct in South America (*Liolaemus lutzae*, *Phymaturus tenebrosus*: $\chi^2 = 32.1$, $P < 0.0001$). In addition, the model predicts recent (2009) extinctions among 24 resurveyed populations of *L. lutzae* ($\chi^2 = 8.8$, $P = 0.003$). In Europe, our

resurvey of *Lacerta vivipara* revealed 14 extinct sites out of 46 (30%), which are predicted quite precisely by the model ($\chi^2 = 24.4$, $P < 0.001$). In Australia, the model pinpoints 2009 extinctions of *Liopholis slateri* ($\chi^2 = 17.8$, $P < 0.00001$) and 2009 extinctions of *Liopholis kintorei* ($\chi^2 = 3.93$, $P = 0.047$). In Africa, analysis of Gerrhosauridae and Cordylidae at 165 sites predicts <1% extinctions, and yet the model pinpoints the single extinction reported by 2009 (exact P -value = 0.006). We temper this value with extinction projections of 23% for 2009 at Malagasy Gerrhosauridae sites, which is validated by the observed 21% levels of local extinction across several lizard families in Madagascar nature reserves (23).

Thermoconforming lizards have been posited (31) to be more vulnerable to climate change relative to heliotherms. Even though $\bar{T}_b$ of thermoconformers ($27.5^\circ\text{C} \pm 1.8^\circ$) is significantly less than $\bar{T}_b$ of heliotherms ($33.5^\circ\text{C} \pm 1.3$, $t = 2.66$, $P < 0.02$, $n = 34$ families; Table 1), PICs show that extinction risk was unrelated to thermoregulatory mode (fig. S8), but was significantly increased by low $\bar{T}_b$, low h_r , and high $\bar{T}_{\max}$. The similar level of local extinctions in 2009 for Malagasy thermoconformers (21%, $n = 63$) and heliotherms [21%, $n = 34$; (23)] supports this view. Evolved changes in thermoregulatory mode, T_b , h_r , lay date, and habitat preference set risk as $T_{\max}$ rises, but owing to trade-offs, T_b and h_r cannot be simultaneously maximized, hence extinction risk is independent of mode (fig. S8). Moreover, extinction risk is not higher for conformers because heliotherms inhabit equatorial regions (i.e., sub-Saharan Africa) that are unavailable to thermoconformers [a factor not considered by (31) or other models (10)], and these areas are warming rapidly (Fig. 3).

Our model, based on T_b , h_r in activity during reproduction, and timing of breeding, assesses salient adaptations that affect thermal extinctions. Concordant verification of 2009 levels of local lizard extinction in North and South America, Europe, Africa, and Australia confirm that extinctions span tropical, temperate, rainforest, and desert habitats. Estimates of evolutionary rates required to keep pace with global change indicate that sustained and intense selection compromises population growth rates, precipitating extinctions. Probability of local extinction is projected to result in species extinction probabilities of 6% by 2050 and 20% by 2080 (table S8). Range shifts only trivially offset losses, because widespread species with high T_b shift to ranges of endemics, thereby accelerating their demise. Although global efforts to reduce CO_2 may avert 2080 scenarios, 2050 projections are unlikely to be avoided; deceleration in $\bar{T}_{\max}$ lags atmospheric CO_2 storage by decades (4). Therefore, our findings indicate that lizards have already crossed a threshold for extinctions.

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- Research of B.S. was funded by the National Geographic Society, UC Mexus, UCSC Committee-On-Research, NSF awards (DEB 0108577, IBN 0213179, LTREB DEB 051597), CNRS fellowships, and visiting professorships (Museum Nationale d'Histoire Naturelle, Université Paris 6, Université Paul Sabatier Toulouse III), PAPIIT-UNAM IN213405 and 224208 to F.M.-C., a Université Paul Sabatier Toulouse III Visiting Professorship to D.B.M., CONACYT grants (4171N and 52852Q) to M.V.-S.C., grant CONACYT-SEP (43142-Q) to H.G., a CONACYT fellowship to R.N.M.-L., CNRS funding to B.H., and M.M., Biodivera: Tenlamas and from ANR Blanche: DIAME to J.C., CONICET grants to L.J.A. and M.M., FONDYCE 1090664 grants to P.V.S., CGL2005-03156 and CLG2008-04164 grant from SMSI to I.J.R., APCT-PICT1086 grant to N.I., scholarships and grants from Universidad Nacional Autónoma de México and American Museum of Natural History to M.V.-S.C., Academy of Finland grant (108955) to T.A.O., Australian Research Council grants to D.G.C., NSF awards DEB 0515909 and 0844523 to A.M.B., NSF award OISE 0530267, PIRE-Patagonia grant to J.W.S., L.J.A., M.M., and P.V.S. and Brigham Young University funding (Biology Department, Kennedy Center for International Studies, Bean Life Science Museum) to J.W.S.

Supporting Online Material

www.sciencemag.org/cgi/content/full/328/5980/894/DC1

Materials and Methods

Figs. S1 to S9

Tables S1 to S8

References

16 November 2009; accepted 7 April 2010

10.1126/science.1184695

Carbon Dioxide Enrichment Inhibits Nitrate Assimilation in Wheat and *Arabidopsis*

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The concentration of carbon dioxide in Earth's atmosphere may double by the end of the 21st century. The response of higher plants to a carbon dioxide doubling often includes a decline in their nitrogen status, but the reasons for this decline have been uncertain. We used five independent methods with wheat and *Arabidopsis* to show that atmospheric carbon dioxide enrichment inhibited the assimilation of nitrate into organic nitrogen compounds. This inhibition may be largely responsible for carbon dioxide acclimation, the decrease in photosynthesis and growth of plants conducting C_3 carbon fixation after long exposures (days to years) to carbon dioxide enrichment. These results suggest that the relative availability of soil ammonium and nitrate to most plants will become increasingly important in determining their productivity as well as their quality as food.

The concentration of CO_2 in Earth's atmosphere has increased from about 280 to 390 $\mu\text{mol CO}_2$ per mol of atmosphere ($\mu\text{mol mol}^{-1}$) since 1800, and predictions are that it will reach between 530 and 970 $\mu\text{mol mol}^{-1}$ by the end of the 21st century (1). Plants could mitigate these changes through photosynthetic conversion of atmospheric CO_2 into carbohydrates and other organic compounds, yet the potential for this mitigation remains uncertain. Photorespiration is the biochemical pathway in which the chloroplast enzyme Rubisco catalyzes the oxidation of the

high-energy substrate RuBP rather than catalyzes the carboxylation of RuBP through the C_3 carbon-fixation pathway (2). Elevated CO_2 (or

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